

Travlytix: Turns to Google Cloud to run a customer data platform and personalization engine

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ABOUT TRAVLYTIX

With big data processing running on Google Platform, Travlytix an intelligent customer data platform for airlines in months with a small engineering team



About Travlytix

Travlytix (<https://travlytix.com/>) provides a customer data platform and personalization engine for airlines and online travel agents. The platform and engine enable airlines to capture and process large amounts of customer data to review behaviors and patterns in real time and take actions accordingly. The Travlytix application empowers marketers, data teams, and senior managers to make data-driven decisions.

The business emerged from an opportunity to

- Creates a customer data platform in six months with a small engineering team
- Provides an always-on, low-latency service to customers
- Innovates faster with big data solutions

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unify and package data for actions and insights. Customer data in the airline ecosystem often resides in fragmented, siloed environments, including booking systems, loyalty systems, and third-party tools. Travlytix resolves customer identities in real time, allowing airlines to communicate more effectively with customers and maintain healthy and profitable relationships.

Industries: Travel & Hospitality

Location: Malaysia

Global travel industry bookings : and, according to analysts, exce year. Travelers generate data ac touchpoints, presenting opportu businesses in the industry to un how they behave and the experie Malaysia-headquartered Travlytix a platform to consolidate real-ti touchpoints into a single locatio for airlines and other industry pa this data to improve profitability customer experience. The busin ensure the data was secured from leakage, disruption or theft.

Platform
(<https://cloud.google.com/ai-platform/>)
Google Cloud (<https://cloud.google.com/>) query/),

App Engine (<https://cloud.google.com/appengine/>)
(<https://cloud.google.com/pubsub/>), Dataflow

AI Platform (<https://cloud.google.com/ai-platform/>)
Memorystore

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)
Cloud Key Management Service

Dataflow (<https://cloud.google.com/dataflow/>)

Engine (<https://cloud.google.com/compute/>)
Cloud Memorystore
(<https://cloud.google.com/memorystore/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Key Management Service
(<https://cloud.google.com/kms/>)

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Compute Engine
(<https://cloud.google.com/compute/>)

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Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

TensorFlow (<https://www.tensorflow.org/>)

"Google Cloud Platform services like Cloud Pub/Sub, Cloud Dataflow, and BigQuery allow us to minimize the effort and resources needed to build data pipelines for airline customers and focus instead on the quality, volume, and velocity of data."

—Srinivas Sri

*Perumbuduru, Head of
Product, Travlytix*

"We come to work every day at Travlytix because we want to solve the biggest problem in travel — data," explains Ron Ramanan, Chief Executive Officer,

Travlytix. "Airlines, online travel agents, and travel ecommerce companies face challenges capturing, processing, and utilizing data to maximize revenue and efficiency. They struggle with fragmented ecosystems, dated technologies, and communicating objectives to multiple vendors and service providers."

The business turned to the cloud to deliver its platform. "Using cloud was a no-brainer. We would not have to reinvent the wheel or work from the bottom up," says Ramanan. "The cloud gave us an opportunity to work with large volumes of data while becoming more agile."

Making data transparent

"Throughout the history of business, people have used data to make more informed decisions," adds Srinivas Sri Perumbuduru, Head of Product at Travlytix. "Our mission at Travelytix is to make data more transparent."

To provide what Sri Perumbuduru describes as "one of the most actionable customer data platforms in the travel industry," Travlytix needed a highly available, resilient, and secure cloud platform. The business turned immediately to [Google Cloud Platform](https://cloud.google.com/) (<https://cloud.google.com/>).

Google Cloud Platform supports architecture principles

Travlytix' analysis found Google Cloud Platform could support its founding principles of keeping its

architecture simple, based on microservices, and highly secure. "Google Cloud Platform documentation and the skills and easy interaction with Google's customer engineers made this possible," explains Sri Perumbuduru. With Google Cloud Platform, the business was able to build its customer data platform in just six months.

"Google Cloud managed services give us more time to strategize our data pipeline architecture and allow us to be less concerned about maintaining the infrastructure."

—Kevin Chin, Data Engineer, Travlytix

Travlytix uses Google Cloud Platform services to collect, process, and store traveler data. The

platform captures data in real time, processes it through data pipelines, and stores it in a BigQuery (<https://cloud.google.com/bigquery/>) data warehouse for analysis. Cloud Pub/Sub (<https://cloud.google.com/pubsub/>) provides the event ingestion and delivery to support the pipelines and Cloud Dataflow (<https://cloud.google.com/dataflow/>) provides the required data processing.

With Google Cloud Platform, Travlytix has been able to break its complex data engineering platform down into small parts, while making data available in real time, running complex data analysis queries in seconds, and scaling seamlessly while taking advantage of on-demand pricing.

"Google Cloud Platform services like Cloud Pub/Sub, Cloud Dataflow, and BigQuery allow us to minimize the effort and resources needed to build data pipelines for airline customers and focus instead on the quality, volume, and velocity of data," says Sri Perumbuduru. "These services enable us to deliver our platform without having to manage hardware or spend weeks on deployments."

"Google Cloud managed services give us more time to strategize our data pipeline architecture and allow us to be less concerned about maintaining the infrastructure," says Kevin Chin, Data Engineer, Travlytix.

Once the data is processed through Google Cloud Platform services, it is sent to Travlytix for airlines and other travel industry participants to undertake marketing automation and analytics.

Travlytix also relies on Compute Engine (<https://cloud.google.com/compute/>) to provide compute resources; Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) to manage and orchestrate Docker containers; App Engine (<https://cloud.google.com/appengine/>) to develop and host its applications; Cloud SQL (<https://cloud.google.com/sql/>) to run its relational databases and Cloud Memorystore (<https://cloud.google.com/memorystore/>) to provide caches to enable sub-millisecond data access.

"Managed containerization through Google Kubernetes Engine and database services through CloudSQL allow us to focus on clients rather than operations," says Sri Perumbuduru.

"Cloud ML Engine enables us to obtain insights from large data volumes in an unbelievably convenient fashion. We use data preparation, model training,

and prediction in
combination to
achieve great
results."

—Rasoul Banaeeyan, Data
Scientist, Travlytix

Low-latency services

Travlytix also uses multiple Google Cloud Platform zones to provide an always-on, low-latency service to clients around the world. Further, solutions like

Cloud Key Management Service

(<https://cloud.google.com/kms/>), which allows the business to manage cryptographic keys and protect sensitive data in Google Cloud Platform, support compliance with security regulations worldwide. These support Travlytix' own encryption of data during collection and processing to protect it from intrusion.

ML provides greater relevance

Machine learning through Google Cloud Platform also gave Travlytix an opportunity to make its products more intuitive and relevant. The business found the TensorFlow (<https://www.tensorflow.org/>) open source library and Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) in particular could help create superior machine learning models. "We preprocess petabyte-scale

records of air travelers' interactions through Google data preparation tools, and all our machine learning models are trained and deployed using its model training and prediction," explains Rasoul Banaeeyan, Data Scientist, Travlytix. "Cloud ML Engine enables us to obtain insights from large data volumes in an unbelievably convenient fashion. We use data preparation, model training, and prediction in combination to achieve great results."

Travlytix can now apply these scalable models to deliver features such as smart customer segmentation and intelligent recommendations that can optimize customer interactions. "With machine learning through Google, we can instantly set up models for prediction and inference," says Banaeeyan. "In addition, BigQuery and TensorFlow help speed the development and rollout of our platform to our customers. We provided direct access to BigQuery to our clients via custom APIs and keep our team in the loop at each stage of development. Most importantly, our customers are updated in real time with the latest results of our analysis."

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